Surface-Prep — Standard Operating Procedure

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> **Audience:** Maintenance Tech · Line Manager **Applies to:** Line-D, station 1, Surface-Prep

1. Purpose

Surface-Prep is the entry of Line-D (Surface Treatment). It uses abrasive blasting / sanding to remove oxide, scale, and contaminants before the part goes into chemical wash and paint. A stoppage here starves every station downstream including the long-cycle Curing-Oven.

2. Normal operating envelope

- Ideal cycle time: **20 s**
- Max acceptable cycle time: **26 s**
- Expected reject rate: **≤ 1 %**
- Fault probability per cycle: **0.3 %**
- Input buffer capacity: **5 parts**

3. Idle reasons

`idle_reason`	Meaning here	First check
`Starved`	Raw part feed empty.	Refill or restart upstream feeder.
`Blocked`	Chemical-Wash (station 2) buffer full.	Verify Chemical-Wash state.

4. Faults & corrective actions

4.1 `Abrasive-Wear`

- **Symptoms in telemetry:** `fault_type = "Abrasive-Wear"`. Surface finish rejects rise at Coating-Inspection.
- **Likely root cause:** blast media degraded; bulk size distribution shifted fine.
- **Corrective action:**

1. Sieve and replenish media to spec.
2. Verify air pressure at the nozzle (target: 90 psi).

- **Restart criteria:** surface roughness matches recipe.

4.2 `Motor-Fault`

- **Symptoms in telemetry:** `fault_type = "Motor-Fault"`. Cell drive motor trips or overheats.
- **Likely root cause:** dust ingress into motor enclosure, bearing wear, or VFD fault.
- **Corrective action:**
  1. LOTO. Inspect motor; clean dust accumulation.
  2. Check VFD fault code; replace fan or motor as indicated.
- **Restart criteria:** stable motor temp / current draw across 10 minutes.

4.3 `Dust-Overload`

- **Symptoms in telemetry:** `fault_type = "Dust-Overload"`. Cell exhaust flow drops; environmental alarms may trip.
- **Likely root cause:** dust collector cartridge plugged.
- **Corrective action:**
  1. LOTO. Pulse-clean or replace the cartridge.
  2. Empty the dust hopper.
  3. Verify exhaust flow against spec.
- **Restart criteria:** static pressure within range.

5. Preventive maintenance schedule

- **Daily:** empty dust collector hopper.
- **Weekly:** check media level and sieve fines.
- **Monthly:** replace dust cartridge.
- **Quarterly:** motor and VFD service.

6. References

- `Maintenance_Workflow_Overview.pdf`
- `Safety_Lockout_Tagout_Procedure.pdf`
- `Line-D_02_Chemical-Wash_SOP.pdf`
- `Production_Lines_Reference.pdf`